

JeongHyeon Kim

M.S. Student, Mechanical Engineering, Korea University
Human-Machine Systems Lab (Advisor: Prof. Shinsuk Park)

+82-10-8274-5411
✉ ahdtd54@korea.ac.kr
🌐 github.com/Juunghyeon
🌐 Juunghyeon.github.io

RESEARCH INTEREST

My research interests lie in **control-oriented dynamics modeling** and **whole-body control** for humanoid robots. I am interested in uncovering the physical relationships that govern robot behavior for a given task and formulating dynamics models around those relationships, so that robots can coordinate full-body motion robustly as contact conditions, payloads, terrain, and external interactions vary.

EDUCATION

- **Korea University (GPA: 3.78/4.0, 4.25/4.5)** Mar.2025 – Feb.2027 (Expected)
M.S. in Mechanical Engineering Seoul, Republic of Korea
- **Korea University (GPA: 3.48/4.0, 3.9/4.5)** 2019 – 2025
B.S. in Mechanical Engineering Seoul, Republic of Korea

EXPERIENCE

- **Human-Machine Systems Lab, Korea University** Mar.2025 – Present
Graduate Research Student (M.S.) Seoul, Republic of Korea
 - **Whole-body Adaptation for Humanoid Standing Manipulation:** formulated the shared physical effect of upper-body motion, payload, and external interaction on whole-body balance as a **Balance Demand Descriptor (BDD)** based on centroidal dynamics, and incorporated it directly into locomotion control so the controller explicitly accounts for changing balance demands
- **Human-Machine Systems Lab, Korea University** Mar.2024 – Feb.2025
Undergraduate Research Student Seoul, Republic of Korea
 - **Visual Perception for Quadrupedal Object Interaction:** designed a dual-camera perception pipeline (global camera for workspace-level observation, local camera for precise near-field sensing) calibrated into a consistent coordinate frame, delivering task-appropriate visual information in real time despite viewpoint changes and occlusion during locomotion
 - **Real-time Architecture for a Generative-AI-based Mobile Manipulator:** designed the real-time communication framework connecting simulation, an LLM, the user interface, and the physical robot; reorganized data flow around the robot's control cycle to resolve cross-module latency and meet real-time constraints

PROJECTS

- **Quadruped Robot Obstacle Avoidance (CCP) — Team Leader** Mar.2024 – Jan.2025
Computer Vision, Autonomous Navigation
 - Developed obstacle detection and avoidance pipeline for DOGZILLA quadruped robot
 - Implemented monocular ORB-SLAM for depth estimation and UV-disparity for obstacle segmentation
- **Ladder-Crossing Robot — Team Leader** Mar.2023 – Sep.2023
Mechanism Design and Control
 - Designed the ladder-crossing mechanism and implemented the Arduino-based control code
- **Solar Tracking Parasol — Team Leader** Sep.2023 – Nov.2023
Mechanism Design and Control
 - Designed the link mechanism and implemented the sun-tracking control code

PUBLICATIONS & MANUSCRIPTS

Manuscripts in Preparation

- J. Kim and S. Park, “Balance Demand Descriptor: A Physics-Informed Representation of Whole-Body Dynamics for Humanoid Standing Manipulation,” *manuscript in preparation*.

Conference Presentations

- J. Kim and S. Park, “Obstacle Detection Using ORB-SLAM and UV-Disparity with a Low-Cost Quadruped Robot,” *KSPE 2025 Spring Conference*, Jeju, Republic of Korea, May 2025.

TEACHING EXPERIENCE

Teaching Assistant, Korea University, Seoul, Republic of Korea

- Robotics Spring 2026
- Computer-Aided Design Fall 2025
- Dynamics (English) Summer 2025
- Kinematics of Mechanisms Spring 2025

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB
Robotics: ROS2, Gazebo, Isaac Gym, MuJoCo, Pinocchio, URDF/XACRO
Control & Optimization: Reinforcement Learning, Trajectory Optimization
Tools: OpenCV, Arduino, Creo, NX, AutoCAD, SolidWorks, Git
OS: Linux (Ubuntu)

HONORS AND AWARDS

Fellowships and Scholarships

- Master’s Research Encouragement Grant, National Research Foundation of Korea 2025
Awarded to selected research-track graduate students for research funding support

Awards

- President’s Award, Creative Challenger Program (CCP), Korea University Jan.2025
Awarded for outstanding performance in autonomous obstacle avoidance for a quadruped robot
- Bronze Award, Creative Comprehensive Design Competition, Korea University Sep.2023
Awarded for the Solar Tracking Parasol mechanism design project
- Design Award, Capstone Design Fair, Korea University Nov.2023